



MULTI-SOURCE LEAD MANAGEMENT & TELECALLING SYSTEM

Product Requirements • Technical Specification • Work Instructions

RECOMMENDED OUTCOME One unified CRM for XpertIntern that captures identifiable student intent from every website and campaign source, assigns leads to managers and agents, enables TeleCMI click-to-call, records every interaction, schedules follow-ups, and reports conversion from first interest to successful enrolment.

Prepared for

XpertIntern • Unit of Xpert Ventures Private Limited

www.xpertintern.com

Version 1.0 • 16 August 2026

Document purpose and recommended decision

This document is the implementation-ready specification for replacing the current basic Leads page with a complete lead management and telecalling system. It is intended for XpertIntern leadership, product/design teams, developers, QA testers, managers and calling agents.

DECISION Build a single canonical lead record with multiple source events, not separate duplicate records for each form or action. Present source-specific views in the interface while preserving one complete student timeline.

Document map

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Document control

Item	Value
Owner	XpertIntern product/operations leadership
Version	1.0
Status	Proposed implementation baseline
Prepared	16 August 2026
Scope	xpertintern.com website, lead CRM, role dashboards and TeleCMI integration
Assumption	The current application codebase and payment gateway remain in place; the new module integrates with them through backend services/events.

1. Current scenario and gaps

The current admin Leads screen already provides a useful starting point: status tabs, search, query type, export, student details and an unassigned state. The Contact Us and Request a Free Call Back pages also collect name, contact number, course, university, semester and a message.

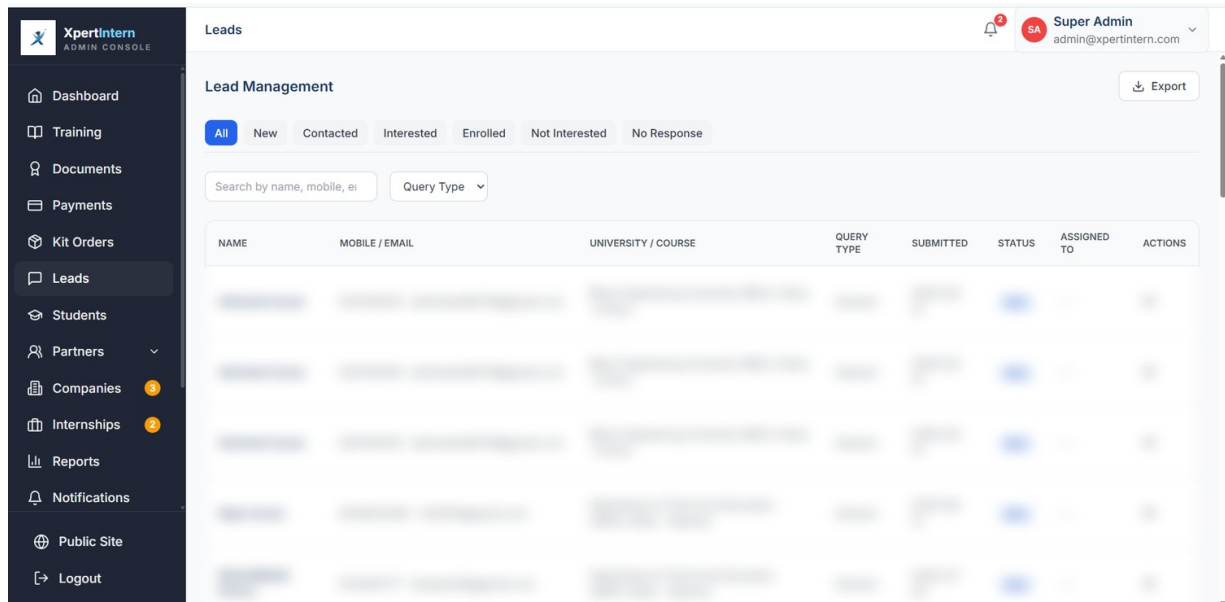


Figure 1. Current Leads page (student row details obscured for privacy).

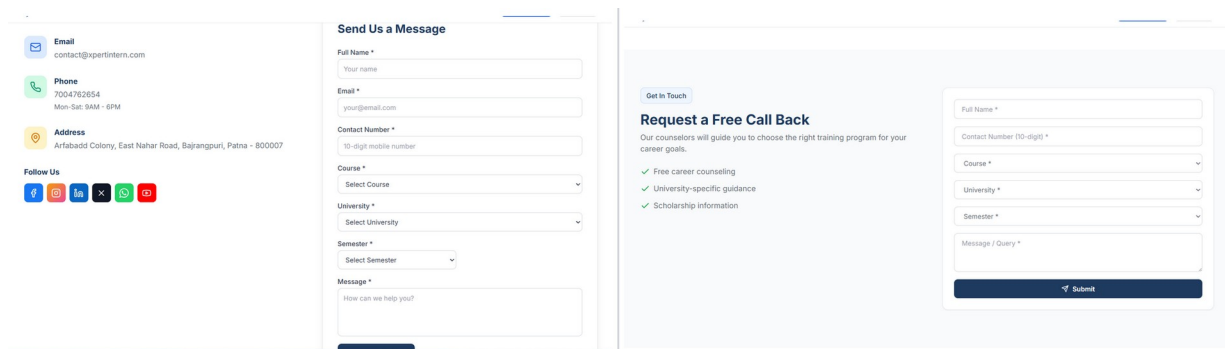


Figure 2. Current Contact Us and Request a Free Call Back capture forms.

Observed gaps

- A form submission appears as a row, but the system does not yet show a complete source history, UTM/campaign context or cross-source identity.
- Repeated submissions can create multiple records for the same mobile number or email, making assignment and conversion reporting unreliable.
- There is no complete team structure for Super Admin, Manager and Agent with strict record-level permissions.
- Lead assignment, capacity balancing, follow-up reminders and reassignment controls are not yet visible.

- Calling happens outside the lead workspace; call UUID, outcome, duration, agent, notes and recording are not automatically tied to the lead timeline.
- The current status set is too broad for day-to-day calling operations and does not enforce a next action after each call.
- Source-specific download, bulk upload validation, duplicate handling, audit history and conversion attribution require a formal design.
- Anonymous training-page clicks cannot safely become a lead until a student is identifiable; otherwise the CRM will fill with unusable records.

2. Target solution and design principles

Recommended system architecture

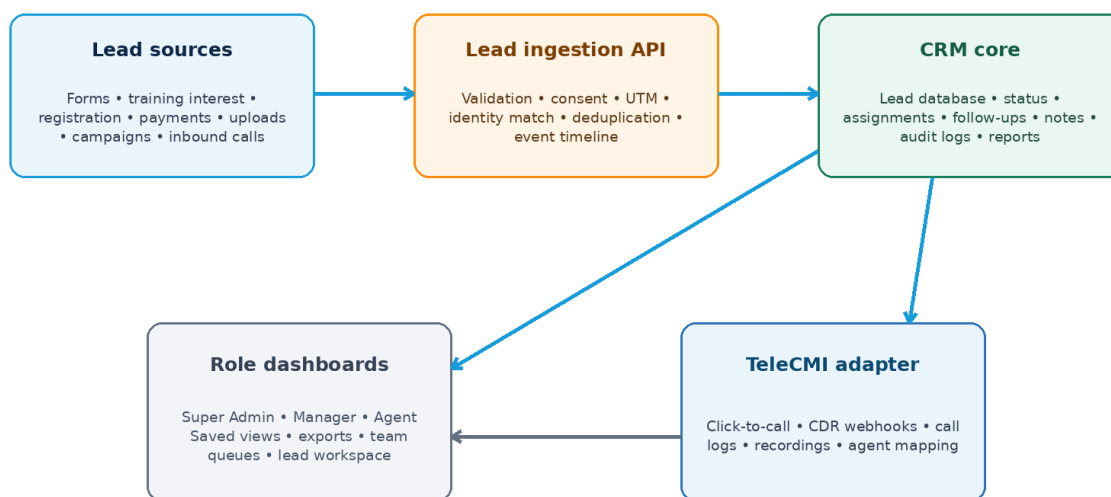


Figure 3. Recommended architecture for lead capture, CRM operations and TeleCMI.

Principle	How it changes the product
One student, one canonical lead	Keep one lead per verified identity and attach every source/action as a timeline event.
Source-aware views	Give separate sections and downloads for Contact, Callback, Training Interest, Registration, Payment Recovery, Campaigns, Inbound Calls, Uploads and Converted leads.
Action-based work queues	Agents should work from Today, Overdue, New Assigned and Follow-up queues instead of scanning a large table.
Telephony inside the lead profile	The Call button, live state, disposition, notes, CDR and recording must form one workflow.
Server-side trust boundary	Website events, payment events, TeleCMI credentials and recording streams must be processed by the backend.

Principle	How it changes the product
Measurable conversion	Report unique leads and progression from source -> contact -> interest -> payment -> enrolled.
Privacy by design	Limit exports and recordings by role, support DND/consent, protect credentials and keep an audit trail.

3. Lead source catalogue and capture rules

The navigation should expose source-specific saved views. Each view uses the same lead database, applies its source/event filter, displays a source count, and supports its own filtered export for authorized roles.

Lead section	Creation trigger	Required context	Default priority
Contact Us	Form submitted	Name, email, mobile, course, university, semester, message	High
Callback Request	Callback form submitted	Name, mobile, course, university, semester, query, preferred time	High
Training Interest	Known user clicks View details / Apply / Request info	Training ID, page, CTA, session, logged-in student or captured contact	Medium
Registration Started	Student submits first identifiable step	Mobile/email, program, university, form step, last activity	High
Incomplete Registration	Started but not completed within configured time	Last completed step, missing fields, abandonment time	High
Payment Created	Payment order/link is created	Order ID, amount, training, coupon/referral, gateway	Very high
Payment Failed/Abandoned	Failure callback or no completion within configured window	Reason, amount, attempts, last gateway event	Very high
Payment Successful	Verified gateway success	Payment ID, amount, training, revenue attribution	Converted
Inbound/Missed Call	TeleCMI CDR for unknown or known number	Direction, agent/team, status, UUID, time	High
Campaign / QR / Referral	Identifiable form/registration action with attribution	UTM, landing page, QR code, coupon, referral partner	Configurable
Manual Upload	Admin/manager imports spreadsheet	Uploaded fields plus batch, uploader and row outcome	Configurable
API / Partner	Authorized system posts lead event	Partner ID, campaign, consent evidence, payload reference	Configurable

Important anonymous-activity rule

DO NOT CREATE ORPHAN LEADS A simple anonymous page view or training-card click is an analytics event, not yet a CRM lead. Store it against a first-party session ID. Convert/merge it into a lead only after the student supplies a mobile/email, logs in, starts registration, requests a callback or creates a payment.

Recommended additional lead-generation points

- Sticky Request Callback button on each training page, prefilled with the training name.
- Exit-intent or delayed enquiry prompt only on high-intent pages; frequency-capped to avoid annoying students.
- WhatsApp click tracking with UTM/campaign code; create a lead only when identity is available or a form is submitted.
- Download syllabus / brochure form with minimal fields (name, mobile, course/university) and clear consent text.
- Payment-recovery lead when an order is created but not verified successfully within the configured period.
- Missed inbound call auto-lead from TeleCMI, with automatic callback SLA and source = Inbound Call.
- Campaign landing pages with UTM source/medium/campaign/content/term, QR ID, influencer/referral ID and coupon code.
- Webinar/orientation registrations and attendance imports linked to the appropriate university, college and training.
- Existing student cross-sell lead only when the student expresses interest in another training; never duplicate the enrolled profile.

4. Lead identity, deduplication, scoring and lifecycle

4.1 Canonical identity and merge rules

- Normalize Indian mobile numbers to E.164 (+91XXXXXXXXXX); remove spaces, leading zero and punctuation before matching.
- Normalize email to lowercase and trim whitespace. Do not merge only on name because names are not unique.
- Primary match: verified/valid normalized mobile. Secondary match: normalized email. If mobile and email point to different existing records, send to a conflict queue for manager review.

- On a match, update only trustworthy newer fields, append the new source event, and preserve the previous values in field history/audit log.
- Never delete the old source. One lead may show Contact Us + Registration Started + Payment Failed + Callback Requested in chronological order.
- Provide Merge Duplicates to Super Admin; Manager may suggest a merge but should not permanently merge without permission.

4.2 Lead score and temperature

Use a configurable score so managers can work high-intent students first. Cap the total at 100 and store an explanation of every scoring component.

Signal	Score	Reason
Payment failed/abandoned	+30	Strong purchase intent
Payment order created	+25	Reached checkout
Callback requested	+25	Asked for human contact
Registration incomplete	+20	Provided identity and started conversion
Contact Us submitted	+20	Direct enquiry
Known training interest	+10	Specific product intent
Repeated high-intent action within 7 days	+10	Recency/frequency
No response after configured attempts	-10	Reduces immediate priority; does not close automatically
Invalid number / DND / explicit no	Close	Remove from active calling queue

Suggested temperature: Hot = 70-100, Warm = 40-69, Cold = 0-39. Keep thresholds configurable by Super Admin.

4.3 Lifecycle

Lead lifecycle and decision flow

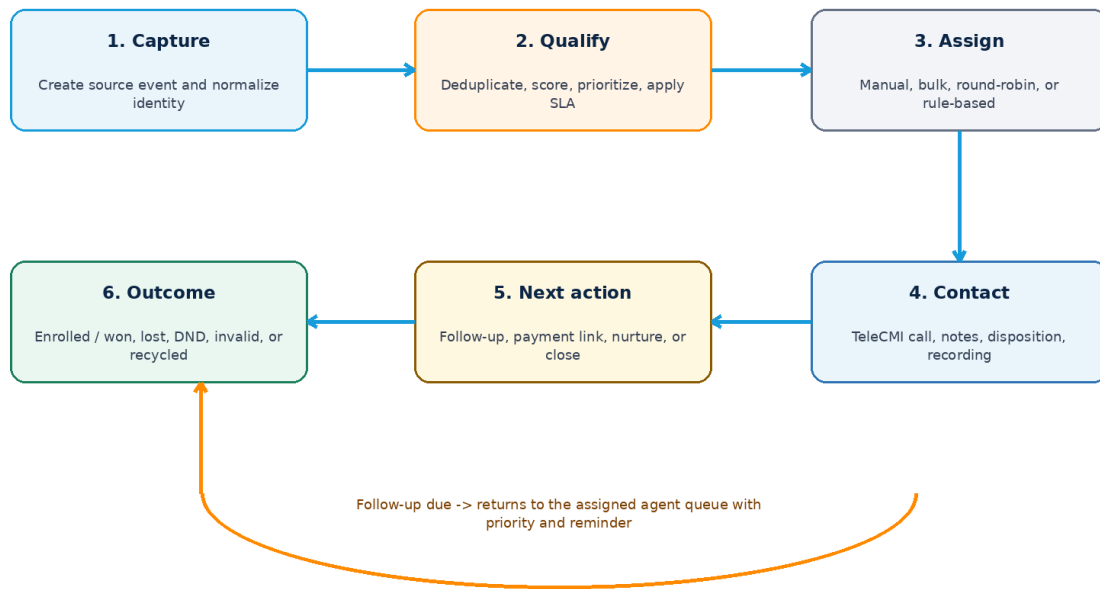


Figure 4. Lead lifecycle with follow-up feedback loop.

Stage	Definition / exit condition
New	Identifiable lead created and not yet assigned/contacted
Assigned	Owned by an active agent and waiting in queue
Attempted	At least one call attempt; student not yet connected
Connected	Two-way conversation occurred
Interested	Student confirms interest; next step must be recorded
Follow-up Scheduled	Specific date/time, reason and owner exist
Payment Link/Order	Payment action created and pending verification
Enrolled / Won	Verified successful payment or authorized enrolment
Not Interested / Lost	Reason required; may be recycled only by manager/admin
No Response	Attempt policy exhausted; may enter nurture/recycle queue
Invalid / Duplicate	Wrong number, duplicate or unusable data
DND / Suppressed	No further promotional calling; reason and source recorded

4.4 Call dispositions

- Interested - payment link requested
- Interested - needs course/university confirmation

- Follow-up - student asked for a specific date/time
- Follow-up - guardian decision / fee discussion
- No answer
- Busy
- Switched off / unreachable
- Wrong number / invalid
- Not interested - price
- Not interested - timing/mode/domain
- Already enrolled elsewhere / already paid
- Do not call / DND
- Escalation to manager

MANDATORY AFTER-CALL RULE A call cannot be marked complete until the agent selects a disposition and enters a meaningful note. Interested and Follow-up dispositions also require a next action and due date/time.

5. Roles, permissions and three dashboards

5.1 Permission model

Capability	Super Admin	Manager	Agent
View leads	All	Team / assigned scope	Own assigned leads
Create/edit lead	Yes	Within team; no protected fields	Limited notes/disposition/contact fields
Assign/reassign	Any user/team	Own team	No
Manage users/roles	Yes	Manage team agents only	No
Call through TeleCMI	Yes	Yes	Assigned leads only
Recordings	All authorized calls	Team calls	Own calls / assigned lead history
Bulk upload	Yes	Yes, if enabled	No
Export/download	All filtered data	Team filtered data	No by default
Merge/delete/archive	Yes with audit	Request/archive within team	No
Configure sources/status/SLA	Yes	No	No
Reports	Organization	Team	Personal only
Audit logs/integration settings	Yes	No	No

Recording access is available to all three roles only within their authorized scope. This satisfies operational access while preventing an agent from browsing recordings belonging to unrelated teams or leads.

5.2 Super Admin dashboard

- Overview cards: new leads, unassigned, overdue follow-ups, calls today, connection rate, interested, payment pending, enrolments and revenue.
- Organization funnel by date and source; source conversion and revenue; university/college/training distribution; lead aging and SLA breaches.
- Lead Inbox with all source sections, advanced filters, saved views, bulk assignment, bulk status, tags, merge, archive and export.
- People & Teams: create manager/agent, deactivate user, map TeleCMI agent ID, set capacity, working hours, leave and team membership.
- Telephony: live/near-live call feed where supported, CDR reconciliation, call logs, recordings, costs if returned, integration health and webhook errors.
- Settings: lead sources, scoring rules, lifecycle/dispositions, assignment rules, SLA, import templates, data retention, consent/DND and audit logs.

5.3 Manager dashboard

- Team summary: available agents, workload, unassigned leads, hot leads, overdue follow-ups, calls/connected/talk time and conversions.
- Assignment board with individual agent capacity, last assignment time, current queue and bulk assign/reassign controls.
- Team queue and source sections limited to the manager's team; filters for source, training, university, stage, priority, age, follow-up due and last call result.
- Manager call access for their team and ability to call leads directly, write notes, schedule follow-ups and escalate/close.
- Coaching/QA: listen to team recordings, tag a call for review, add a manager-only QA note and track disposition quality.
- Exceptions: agents inactive/on leave, leads untouched beyond SLA, repeated no-answer, invalid numbers, webhook mismatch and uncompleted after-call work.

5.4 Agent dashboard

- My Day cards: calls due, overdue follow-ups, new assignments, completed calls, connected calls and interested leads.
- Priority queue order: overdue follow-ups -> today's follow-ups -> hot newly assigned -> other assigned leads, subject to calling-window rules.

- Lead workspace shows student summary, training/university/semester, source history, previous calls/notes, next action, Call button and disposition form.
- Agent may update call notes, disposition, follow-up, interest/training selection and limited contact corrections; cannot assign, export, delete or open other agents' leads.
- Phone numbers may be masked in the interface while TeleCMI connects the call, depending on XpertIntern's privacy policy.
- Responsive layout for desktop and mobile; no need to copy numbers or switch tabs for normal calling work.

6. Core modules and screen requirements

Module	Minimum required functions
Lead Inbox	Source sections, status tabs, search, filters, saved views, counts, bulk actions and filtered export
Lead Profile	Identity, source history, interests, status, score, assignment, timeline, calls, notes, follow-ups, payments and audit
Assignment Center	Manual/bulk/rule-based assignment, capacity, round-robin, reassignment reason and history
Call Workspace	TeleCMI Call button, live state, timer, script/context, autosaved notes, disposition, follow-up and recording
Follow-up Center	Today, overdue, upcoming, completed; calendar/list; reminders and ownership
Imports	Template, preview, validation, dedup choice, assignment, progress, error file and batch rollback/archive
Exports	Exact filtered result, selected columns, CSV/XLSX, queued generation, audit and secure expiry
People & Teams	Roles, manager-agent hierarchy, TeleCMI mapping, capacity, active status and working hours
Reports	Funnel, source, agent/team, activity, SLA, aging, payment recovery, training demand and recordings QA
Settings & Audit	Sources, statuses, dispositions, score, SLA, integration, consent, DND, retention and immutable activity log

6.1 Lead list columns

Default columns should be compact and configurable: Name; masked/full Mobile; Email; Source (primary + latest); Training Interest; University/College; Score/Temperature; Stage; Assigned Agent; Last Call Result; Next Follow-up; Age/SLA; Created; Actions. Users may choose columns; the server must enforce role permissions regardless of hidden/visible columns.

6.2 Lead profile layout

- Header: name, mobile/email, stage, temperature, assigned agent, source badges, SLA indicator and Call button.

- Overview: university, college, course/branch, semester, location, preferred language, training interest and consent/DND.
- Timeline: form events, training interest, registration steps, payment events, assignments, calls, notes, follow-ups, status changes and system automation.
- Tabs: Activity; Calls & Recordings; Notes; Follow-ups; Payments; Source & Campaign; Audit (admin only).
- Quick actions: Call, Add note, Schedule follow-up, Send payment link (when available), Change stage, Escalate, Reassign (manager/admin).

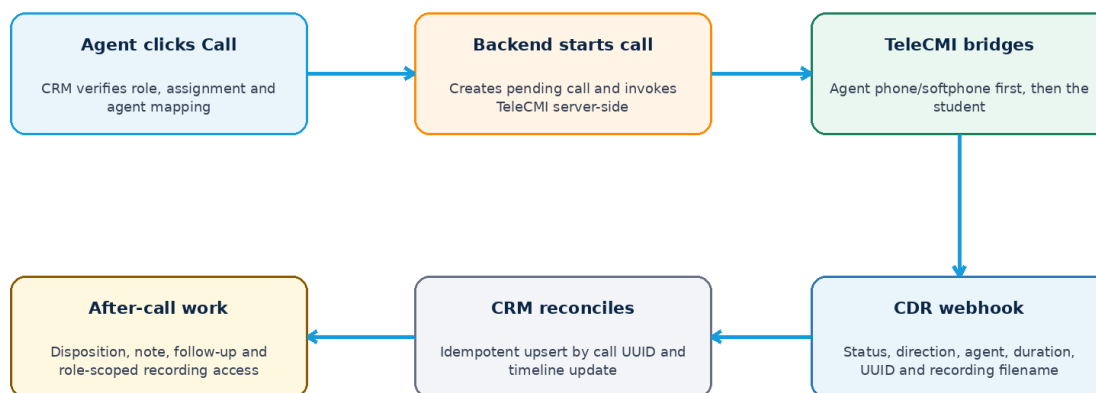
6.3 Filters and saved views

Source and source event; stage/disposition; assigned agent/manager/team; unassigned; priority/score; training/domain; university/college/course/branch/semester; campaign/UTM/QR/coupon/referral; payment state; created/updated/last call/follow-up date; lead age; SLA; tags; consent/DND; call count; connected/not connected; has recording; import batch.

EXPORT BEHAVIOUR Export must use the exact active filters and selected columns. The export screen must show the number of rows, role scope and a data-privacy warning before generation.

7. TeleCMI integration specification

TeleCMI click-to-call and call-log flow



Security boundary: API secrets and recording credentials stay in the backend; the browser receives only authorized actions or short-lived streams.

Figure 5. Recommended click-to-call, CDR and recording workflow.

7.1 Feasibility

Direct calling from the XpertIntern lead profile is feasible. TeleCMI's official documentation describes click-to-call using a user/agent token, an administrative click-to-call option using app credentials plus

agent ID, completed-call webhooks containing CDR data, user call-history APIs and recording playback using the returned filename.

ACCOUNT VERIFICATION REQUIRED TeleCMI publishes more than one endpoint family (including rest.telecml.com/v2 and CHUB/India piopiy endpoints). Before development, XpertIntern must obtain the exact base URL, authentication mode, caller ID/virtual number, recording availability and webhook configuration enabled for its subscription. Implement a TeleCMI adapter so endpoints can change without rewriting the CRM.

7.2 Recommended authentication approach

Choice	Method	Control
Preferred	Server-side admin click-to-call mapped to TeleCMI agent ID	Avoids exposing app secret or student number to the browser; keeps agent attribution when supported
Alternative	Per-agent TeleCMI user token	Use encrypted token storage and refresh/re-authentication; official docs state login tokens may expire
Never	Direct browser call to TeleCMI with app secret	Would expose credentials and allow unauthorized calls

7.3 Outbound call sequence

1. Agent opens an assigned lead and clicks Call.
2. Backend verifies role, assignment, DND, allowed calling time, active TeleCMI mapping and rate limits.
3. Backend creates call_attempt with status = initiating and a correlation ID. Include lead ID/call attempt ID in TeleCMI custom/extra parameters where the account supports it.
4. Backend invokes TeleCMI. TeleCMI connects the agent phone/softphone first and then the student, as described in its click-to-call documentation.
5. UI shows Initiating -> Agent ringing -> Student ringing -> Connected -> Completed where live events are available; otherwise show Initiated and reconcile at CDR receipt.
6. TeleCMI sends completed-call CDR webhook. Backend validates, stores raw payload safely, upserts by cmiuuid/conversation UUID and links agent/lead.
7. CRM updates direction, status, start time, duration/answered seconds, hang-up reason, agent/team, recording flag and filename.
8. Agent completes disposition, note and follow-up. If no manual result is entered, the lead remains in After-call work and manager sees an exception.

7.4 Webhook and reconciliation controls

- Public HTTPS endpoint dedicated to TeleCMI CDRs; verify app ID and payload shape. If TeleCMI supports signature/HMAC or IP allowlisting for the account, enable it.

- Idempotency key = cmiuuid/conversation UUID plus event type; repeated webhooks must not create duplicate calls.
- Use a secret, unguessable webhook path or gateway token if signed webhooks are not available; rate-limit and log rejected requests without logging credentials.
- Store webhook receipt time and processing status. Retry processing internally; return a fast success response after safe persistence.
- Run a scheduled reconciliation job against TeleCMI call-history APIs for recent calls to repair missed webhooks.
- Map TeleCMI user/agent ID to one active CRM user; unknown mappings go to an Integration Exceptions queue.

7.5 Recording access

- Store recording_available and recording_filename/reference, not TeleCMI app credentials or a permanently public recording URL.
- When authorized user clicks Play, backend verifies scope, requests/streams the audio using server-side TeleCMI credentials, and logs the access.
- Super Admin: organization scope; Manager: team scope; Agent: own calls and assigned lead history only.
- Show call date, agent, direction, duration and disposition beside the player. Add playback speed and a manager-only QA note/bookmark later if needed.
- Retention, download permission and recording consent notice must be configurable and reviewed against applicable law and TeleCMI plan terms.

7.6 Minimum TeleCMI data stored

Group	Fields
Identity	cmiuuid/conversation_uuid, CRM call_attempt_id, lead_id, CRM agent_id, TeleCMI agent/user ID
Call	direction, status, from/to (encrypted or protected), virtual/caller ID, team, start/answer/end timestamps
Metrics	duration, answered seconds/billed seconds, hang-up reason, cost/rate only if returned and needed
Recording	record flag, filename/reference, availability, last verified time; never app secret
Payload	raw webhook JSON in protected integration log with retention and access control

8. Lead upload, assignment, download and data quality

8.1 Spreadsheet upload

Category	Template columns
Required	name, mobile
Recommended	email, university, college, course, branch, semester, training/domain, city/state
Attribution	source, campaign, UTM, QR/referral/coupon, event date
Operations	priority, tags, manager/team, assigned agent email/ID, next follow-up, notes
Compliance	consent status/source/time, DND flag where applicable

1. Download the latest template from the system; do not accept arbitrary column names without mapping.
2. Upload CSV/XLSX. Show preview, detected columns, batch name, source and default assignment before import.
3. Validate mobile/email formats, required fields, university/training IDs, dates and agent IDs. Reject formulas/macros and unsafe file types.
4. Show row results: New, Existing matched, Conflict, Invalid or DND. Let authorized user choose Skip, Merge event/update allowed fields, or Create only for verified non-duplicates.
5. Commit in background, show progress and counts, produce an error file for rejected rows, and keep batch audit/rollback-as-archive controls.

Recommended first release limit: 10,000 rows per file, configurable after performance testing. Large jobs must run asynchronously so the dashboard remains usable.

8.2 Assignment methods

Method	Logic	Best use
Manual single	Manager/admin selects one agent	Escalation, special language/domain or VIP lead
Bulk	Select filtered leads and one/team of agents	Campaign or uploaded batch
Round-robin	Distribute among active agents under capacity	Normal inbound flow
Rule-based	Source/university/training/language/region -> team/agent	Specialized teams
Load-balanced	Assign to agent with lowest active weighted queue	Uneven follow-up workload
Sticky owner	Return repeat lead to previous active agent	Continuity; manager can override

Every assignment/reassignment must store from user, to user, reason, timestamp and assignment rule/batch. Deactivated/on-leave agents must not receive new leads; managers receive a reassignment warning for their open queue.

8.3 Download/export

- Admin can export organization data; Manager can export only team data; Agent export is disabled by default.
- Source pages include their own Download button, plus date range, status, assigned agent, training/university and custom field selection.
- Export the current filtered dataset - not the whole table silently - and include export generated time, filter summary and source section.
- For large exports, generate asynchronously and provide a short-lived authenticated download. Record who exported what and when.
- Support CSV and XLSX. Use safe cell escaping to prevent spreadsheet formula injection. Optionally mask phone/email for non-admin exports.

9. Reports, KPIs, alerts and automations

Report	Measures
Lead funnel	Unique leads -> assigned -> attempted -> connected -> interested -> payment created -> enrolled
Source performance	Unique leads, duplicates, contact rate, interest rate, enrolment rate, revenue and cost if available
Agent/team	Assigned, attempts, connected, talk time, after-call completion, follow-up adherence, interested, won and QA
SLA/aging	Unassigned age, first-contact time, overdue follow-ups, untouched leads and stage aging
Payment recovery	Orders created, failed/abandoned, contacted, recovered payments, amount and agent/source attribution
Training demand	Interest and conversion by training/domain, university, college, course/branch and semester
Call quality	No-answer/busy/invalid mix, short calls, recording availability, disposition completeness and manager QA flags
Data quality	Duplicate/conflict rate, invalid numbers/emails, missing source, unknown agent mappings and webhook failures

Recommended operational metrics

Use quality and conversion together; do not rank agents only by raw call count.

KPI	Definition
First contact time	Lead created -> first attempted call
Connection rate	Connected calls ÷ outbound attempts
Contact rate	Unique leads connected ÷ unique leads attempted
Interest rate	Interested unique leads ÷ connected unique leads
Conversion rate	Enrolled unique leads ÷ eligible unique leads

KPI	Definition
Follow-up adherence	Completed on-time follow-ups ÷ due follow-ups
Payment recovery	Recovered successful payments ÷ payment failed/abandoned leads
Disposition completeness	Calls with valid disposition/note ÷ completed calls

Alerts and automations

- New hot lead unassigned beyond the configured SLA -> manager alert; escalate to admin if still unassigned.
- Follow-up due in 15 minutes and overdue -> in-app notification; daily summary to agent/manager.
- Payment failure/abandonment -> high-priority recovery queue and optional approved message/payment-link workflow.
- Missed inbound TeleCMI call -> create/match lead and schedule callback task.
- Agent inactive/on leave -> stop assignment and prompt manager to reassign open follow-ups.
- Webhook/recording/authentication failure -> integration health alert to admin; no silent data loss.
- Repeated invalid/no-response result -> lower priority or recycle after configurable attempt policy; never auto-DND without explicit evidence.

10. Data model, APIs, security and non-functional requirements

10.1 Core entities

Entity	Purpose / key data
users / teams	Role, manager relationship, active status, capacity, working hours, TeleCMI mapping
leads	Canonical identity, profile, stage, score, owner, SLA, consent/DND and latest summary fields
lead_source_events	Every form/action/campaign/payment/registration/inbound-call source event and raw reference
lead_assignments	From/to, rule/batch, reason, assigned_at, ended_at
lead_status_history	Previous/new stage, disposition, actor, reason and time
lead_notes	Call/general note, author, visibility, timestamps and edit history
followups	Owner, due date/time, reason, status, completion and linked call
call_attempts	Lead, agent, provider, direction, UUID, state, times, metrics, disposition and recording reference
payment_events	Order/payment IDs, amount, gateway state, training, coupon/referral and verified timestamps

Entity	Purpose / key data
imports / import_rows	Batch metadata, row validation/match/result and error reason
tags / lead_tags	Flexible operational segmentation
audit_logs / export_jobs	Sensitive actions, before/after metadata, export scope and secure file expiry

10.2 Recommended internal API surface

Endpoint	Responsibility
POST /api/lead-events	Server-side website/payment/registration event ingestion with idempotency key
GET /api/leads	Role-scoped list with source, stage, agent, date and advanced filters
GET/PATCH /api/leads/{id}	Lead profile and permitted updates
POST /api/leads/{id}/assign	Manual/reassignment with reason and authorization
POST /api/leads/bulk-assign	Bulk/rule assignment
POST /api/leads/{id}/call	Create call attempt and invoke TeleCMI adapter
POST /api/webhooks/telecml/cdr	Receive and idempotently process completed call CDR
GET /api/calls/{id}/recording	Authorized short-lived/streamed recording access
POST /api/leads/{id}/disposition	Disposition, note, stage and next action
POST /api/followups	Create/update/complete follow-up
POST /api/imports	Validate/preview/commit uploaded lead batch
POST /api/exports	Create audited filtered export job
GET /api/reports/*	Role-scoped funnel/source/team/agent/SLA reports

10.3 Lead-event example

```
{
  "event_id": "evt_website_01H...", "event_type": "payment.failed",
  "occurred_at": "2026-08-16T05:40:00Z",
  "identity": {"mobile": "+91XXXXXXXXXX", "email": "student@example.com"},
  "profile": {"name": "Student Name", "university_id": "beu", "semester": 3},
  "context": {"training_id": "web-development", "payment_order_id": "order...", "amount": 599},
  "attribution": {"utm_source": "youtube", "utm_campaign": "beu_sep_2026", "coupon": "FLAT100"},
  "consent": {"status": "captured", "source": "checkout"}
}
```

10.4 Security, privacy and audit

- Role-based access on every backend query/action; never rely only on hidden frontend buttons.
- Encrypt sensitive credentials and provider tokens at rest; use a secrets manager/environment configuration, rotation and least privilege.
- Protect mobile/email and recording references; use TLS in transit, database access controls, masked logs and restricted backups.

- Multi-factor authentication for Super Admin/Manager is strongly recommended; session timeout and device/login audit for all roles.
- DND/suppression flag blocks normal call initiation and bulk campaign assignment. Override, if legally permitted, must be admin-only and audited.
- Record all assignment, status, note edit, merge, export, recording access, integration setting and user-management actions.
- Define configurable retention for raw webhook payloads, exports, recordings and inactive lead data; obtain legal review for consent/recording/telemarketing obligations.
- Back up the database, test restore, monitor webhook failures, and provide a recovery playbook for provider outages.

10.5 Non-functional targets

Area	Target / requirement
Performance	Typical list/filter API p95 under 800 ms excluding third-party calls; paginated server-side queries
Availability	Target 99.5% monthly for CRM module, excluding planned maintenance and TeleCMI/provider outages
Scalability	Background queues for imports, exports, webhooks and reconciliation; indexes on normalized identity, owner, stage, source and due time
Reliability	Idempotent website/payment events and TeleCMI CDRs; retries with dead-letter/error queue
Accessibility	Keyboard usable, visible focus, readable status labels, responsive at common desktop/mobile widths
Observability	Structured logs, metrics and alerts for ingestion, assignment, click-to-call, webhook processing, recording and export jobs
Compatibility	Current supported Chrome/Edge plus responsive agent use; provider adapter prevents hard dependency on one API version

11. Work instructions

11.1 Super Admin - initial setup

1. Create Managers and Agents; attach each agent to one manager/team; set active status, capacity, working hours and permissions.
2. Obtain TeleCMI app/account details from TeleCMI, confirm exact API family and recording/webhook availability, then add credentials only in protected server configuration.
3. Map every CRM agent to the correct TeleCMI agent/user ID; run a test call and confirm agent attribution, CDR and recording.
4. Configure lead stages, dispositions, source views, scoring, SLA, assignment rules, attempt policy, DND and retention.

5. Connect Contact Us, Callback, training CTAs, registration, payment gateway events and campaign attribution to the lead-event endpoint.
6. Upload any opening lead batches using preview/validation; resolve conflicts and assign managers/agents.
7. Run UAT with one manager and two agents before enabling all teams; verify permissions using separate test accounts.

11.2 Super Admin - daily/weekly operation

- Review unassigned hot leads, SLA breaches, integration errors and unusual duplicate/conflict rates daily.
- Review source conversion, payment recovery, workload and follow-up adherence weekly; change rules only with a recorded reason.
- Sample recordings and audit exports/recording access. Deactivate departed users immediately and reassign their open work.
- Use archive rather than hard delete for normal cleanup. Preserve source, call and status history.

11.3 Manager - start of day

1. Open Team Dashboard and check available agents, overdue follow-ups, unassigned leads and hot/new queue.
2. Confirm agents are active and not on leave; update capacity or reassign open work where required.
3. Assign leads manually/bulk or review the round-robin result. Prioritize callback, payment recovery and registration-incomplete sources.
4. Review yesterday's incomplete after-call work, SLA breaches and exceptions before distributing additional leads.
5. Share daily focus by training/campaign and confirm approved call script/offer information.

11.4 Manager - during/end of day

- Monitor queue and follow-up completion without reassigning active conversations unnecessarily.
- Call/escalate when needed; add clear notes and ownership. Review selected recordings for coaching and compliance.
- Rebalance leads when an agent becomes unavailable. Always enter a reassignment reason.
- At end of day, confirm overdue follow-ups, unworked hot leads, agent after-call completeness and next-day schedule.

11.5 Agent - standard calling workflow

1. Log in and open My Day. Work overdue/today follow-ups before new assigned leads unless the manager gives another priority.
2. Open the lead, review source, training/university, previous timeline, payment/registration state and last notes before calling.
3. Click Call. Do not copy the student number to a personal dialer for normal work. Wait for TeleCMI to connect the agent leg and student leg.
4. During the call, enter concise notes: need, training discussed, objection/question, offer/payment information and promised next step. Notes should autosave as draft.
5. After the call, select the correct disposition. If Interested or Follow-up, set the next action, date/time and reason. If DND/wrong number, choose that exact result.
6. Confirm Save & Complete. The lead moves to the next correct queue; do not leave completed calls in after-call work.
7. For a due follow-up, read the prior note first. Complete or reschedule only with a reason. Escalate pricing/policy exceptions to the manager.

11.6 Note-writing standard

GOOD NOTE Student wants Web Development offline program for BEU 3rd semester. Asked about start date and ₹599 fee. Explained schedule and sent approved payment link. Student will confirm with guardian; follow up 17 Aug at 5:30 PM.

POOR NOTE Talked / call later / interested. This does not state what was discussed, the objection, the promise or the next action.

12. Implementation roadmap, testing and acceptance

12.1 Recommended phased delivery

Phase	Deliverables
Phase 0 - Discovery	Confirm current stack/database/payment events, TeleCMI account endpoints, roles, data fields, consent text and migration sample
Phase 1 - CRM foundation	Canonical lead, source events, deduplication, lead list/profile, lifecycle, RBAC, teams, manual assignment, notes/follow-ups
Phase 2 - TeleCMI	Agent mapping, click-to-call, CDR webhook, reconciliation, call timeline, dispositions and recording access
Phase 3 - Capture and recovery	Contact/callback, training/registration/payment events, campaign attribution, missed-call lead, source views
Phase 4 - Bulk and reporting	Upload preview/dedup, downloads, dashboards, funnel/source/team reports, SLA/alerts and audit

Phase	Deliverables
Phase 5 - Optimization	Auto-assignment, lead scoring tuning, payment-link actions, QA workflow, approved messaging and advanced analytics

A realistic first production release usually requires multiple development and QA sprints. The schedule depends on the existing xpertintern.com codebase, payment gateway integration, TeleCMI account configuration and data migration quality.

12.2 UAT scenarios

Area	Acceptance test
Source capture	Submit Contact and Callback forms; verify one lead and two source events on repeat identity
Training interest	Known user clicks a training CTA; event joins the lead; anonymous view does not create orphan lead
Registration/payment	Start registration, create/fail/succeed payment; stages and timeline update idempotently
Dedup	Same mobile with different email; email conflict; upload duplicate; verify merge/conflict policy
RBAC	Agent cannot open other agent lead/export; manager limited to team; admin all
Assignment	Manual, bulk, round-robin, inactive/over-capacity agent and reassignment history
TeleCMI call	Agent click-to-call, correct TeleCMI user, correct lead correlation and no secret in browser/network logs
CDR	Answered/missed/repeated webhook; one call record with correct status, duration and agent
Recording	Authorized playback works; unauthorized role blocked; access audited
After-call	Disposition/note required; interested requires next action; DND blocks future normal call
Follow-up	Reminder, overdue queue, reschedule reason and manager exception
Upload/export	Invalid rows, preview, error file, dedup choice, filtered export and formula-injection-safe cells
Failure recovery	TeleCMI timeout/webhook outage; CRM remains usable; reconciliation repairs call log

12.3 Release acceptance criteria

- All defined identifiable website/payment/TeleCMI sources create or update a lead without unintended duplicates.
- Every lead shows source history, owner, stage, call/note/follow-up timeline and conversion/payment outcome where applicable.
- Super Admin, Manager and Agent permissions pass positive and negative tests at API and UI level.
- Manager can assign/reassign leads and see team workload; Agent sees only assigned work and can complete the full call workflow.
- TeleCMI click-to-call initiates from the lead profile; completed CDR correctly maps to lead and agent; repeated webhook is idempotent.

- Recordings play only through authorized role scope without exposing TeleCMI secret; recording access is audited.
- Imports validate and explain every row; exports match filters and role scope; sensitive actions appear in audit logs.
- Overdue follow-up/SLA and integration failures are visible and actionable; no silent loss of call or lead events.
- Dashboards use unique leads for funnel metrics and clearly distinguish call attempts from unique students.
- Production monitoring, backup/restore procedure and TeleCMI outage/reconciliation runbook are approved before go-live.

12.4 Open decisions before development

Decision	Required answer
Technology	Current frontend/backend/database, authentication and hosting details
TeleCMI	Exact plan/API family, app ID/secret handling, agent IDs, caller ID, webhooks, live events and recording retention
Payments	Gateway and authoritative success/failure/order-created webhooks
Identity	Whether one mobile can legitimately represent siblings/guardian and how such conflicts are handled
Calling policy	Allowed hours, attempt limit, DND process, recording notice and retention approved by legal/operations
Messaging	Whether payment links and follow-up messages are sent through existing WhatsApp/SMS/email provider in Phase 1 or later
Migration	How current lead rows are deduplicated and whether historic notes/status/calls exist
Agent privacy	Whether agents see full phone/email or use masked click-to-call only

Appendix A - TeleCMI official references

The following official TeleCMI pages were reviewed on 16 August 2026. Endpoint and entitlement details must be verified against XpertIntern's active TeleCMI account before production.

- [Click-To-Call \(REST v2\)](#)
- [Click-to-call Admin \(India/CHUB\)](#)
- [User/Agent Login Token](#)
- [Webhooks Overview](#)
- [Incoming Answered CDR/Webhook](#)
- [User Outgoing Calls API](#)
- [Download/Play Recorded File](#)
- [Browser/Live Feed SDK](#)

Appendix B - Recommended lead upload header

name, mobile, email, university, college, course, branch, semester, training, city, state, source, campaign, utm_source, utm_medium, utm_campaign, qr_id, referral_id, coupon, priority, tags, assigned_agent, next_followup_at, consent_status, consent_source, event_date, notes

Appendix C - Developer handoff checklist

- ☐ Current schema and code paths for contact/callback/registration/payment reviewed
- ☐ Event types and idempotency keys agreed
- ☐ Canonical identity and conflict policy approved
- ☐ RBAC matrix implemented in backend policies/tests
- ☐ TeleCMI account endpoints and credentials tested in non-production
- ☐ Agent mapping and custom correlation parameter confirmed
- ☐ Webhook security, idempotency and reconciliation implemented
- ☐ Recording proxy and authorization tests complete
- ☐ Import/export security and audit complete
- ☐ Dashboards validated against unique-lead metric definitions
- ☐ Monitoring, alerts, backup/restore and outage runbook complete
- ☐ UAT sign-off by Super Admin, Manager and Agent representatives